M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

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Date:	10 June 2026

1. Introduction

This report presents the work completed for **Task 1** of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Artificial Intelligence and Machine Learning, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

The primary objectives of this task were:

1. **Learn GUI Development**: Gain hands-on experience with Python's Tkinter library for creating desktop applications.
2. **Implement CRUD Operations**: Develop functionality to Create, Read, Update, and Delete building data.
3. **Understand Data Management**: Work with Python data structures (lists, dictionaries) to store and manage project information.
4. **Practice Problem-Solving**: Apply programming logic to solve a real-world campus planning scenario.
5. **Build a Functional Application**: Create a complete, working application with proper user interface and functionality.
6. **Learn Software Design**: Understand how to structure code into organized, manageable functions.
7. **Develop Documentation Skills**: Learn to document code and create user-friendly interfaces.

3. Task Description

Core Requirements:

1. **Building Management**
   - Add new buildings with details (name, area, floors, type, cost)
   - View all buildings in a tabular format
   - Delete individual buildings
   - Clear all building data
2. **User Interface**
   - Clean and professional GUI using Tkinter
   - Left panel for data input
   - Right panel for data display

- Interactive buttons for all operations

3. **Data Features**
    - Display summary statistics (total buildings, total cost)
    - Generate detailed reports
    - Export data to CSV format
    - Pre-loaded sample data for demonstration
4. **Technical Requirements**
    - Use Python 3.x
    - Implement using Tkinter library
    - No external dependencies
    - Error handling and validation
    - User-friendly messages and confirmations

4. Work Done

1. Project Setup and Planning

The project began with setting up the development environment and planning the application structure. The following decisions were made:

- **Technology Choice**: Python with Tkinter for GUI (no external dependencies)
- **Data Structure**: List of dictionaries to store building information
- **Layout**: Two-panel design (input on left, display on right)
- **Features**: CRUD operations, reporting, and data export

2. User Interface Design

The interface was designed with simplicity and functionality in mind:

Input Panel (Left Side):

- Building Name (text input)
- Area in square feet (numeric input, default: 5000)
- Number of Floors (numeric input, default: 2)
- Building Type (dropdown: Academic, Admin, Residential, Lab, Library)
- Construction Cost (numeric input, default: 100000)
- Action Buttons: Add Building, Delete Selected, Clear All

Display Panel (Right Side):

- Treeview widget showing building data in columns
- Columns: ID, Name, Area, Floors, Type, Cost
- Scrollbar for easy navigation

Bottom Panel:

- Summary statistics (total buildings and total cost)
- Show Report button

- Export Data button

4. Data Management

Sample data was added to demonstrate the application's capabilities:

Building Name	Area (sq ft)	Floors	Type	Cost ($)
Science Building	15,000	4	Lab	2,500,000
Library	12,000	3	Library	1,800,000
Admin Block	8,000	2	Admin	1,200,000
Student Center	10,000	2	Academic	1,500,000
Research Lab	20,000	5	Lab	3,500,000

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5. Challenges Faced

1. GUI Layout Management

Challenge: Positioning widgets properly using Tkinter's geometry managers.
Solution: Used a combination of `pack()` and `grid()` methods with proper frames. Left panel used `grid()` for organized input fields, while right panel used `pack()` for the treeview.

2. Data Validation

Challenge: Ensuring users enter correct data types (numbers for area, floors, cost).
Solution: Implemented try-except blocks with specific error messages for each input field.

3. Treeview Widget Configuration

Challenge: Setting up and configuring the Treeview widget with proper columns and formatting.
Solution: Researched Treeview documentation and configured columns with proper widths and headings.

4. Real-time Updates

Challenge: Keeping the display synchronized with data changes.
Solution: Created a `refresh_list()` method that updates the treeview and summary statistics after any data modification.

5. Export Functionality

Challenge: Converting data to CSV format properly.
Solution: Used Python's built-in file handling to write CSV data with proper formatting.

6. User Experience

Challenge: Making the application intuitive and user-friendly.
Solution: Added confirmation dialogs, success messages, and clear button labels.

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6. Key Learnings

Technical Skills Acquired:

1. **Python GUI Development**
   - Understanding Tkinter widgets (Label, Entry, Button, Treeview, Combobox)
   - Layout management using pack() and grid()
   - Event handling and callback functions
   - Message boxes for user interaction
2. **Data Structures**
   - Working with lists and dictionaries

  - Data persistence using CSV files
  - CRUD operations implementation
3. **Software Design Principles**
  - Class-based design for better organization
  - Separation of concerns (UI vs. Logic)
  - Modular functions for maintainability
4. **Problem-Solving**
  - Breaking down complex requirements into smaller tasks
  - Debugging GUI applications
  - Input validation techniques
5. **Project Management**
  - Planning application structure before coding
  - Testing each feature incrementally
  - Documenting code and functionality

Professional Skills Developed:

1. **Time Management**: Completing the task within the given timeframe
2. **Documentation**: Creating clear, organized code with proper comments
3. **User-Centered Design**: Focusing on usability and user experience
4. **Attention to Detail**: Ensuring data accuracy and proper formatting

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7. Conclusion

The Campus Expansion Planner project successfully demonstrates the application of Python programming skills to a real-world scenario. The application provides a practical solution for educational institutions to plan and manage their campus expansion projects efficiently.

Intern Signature:	________________________	**Supervisor:**	________________________
Name:	Intern Name	**Name:**	
Date:	June 2026	**Designation:**	